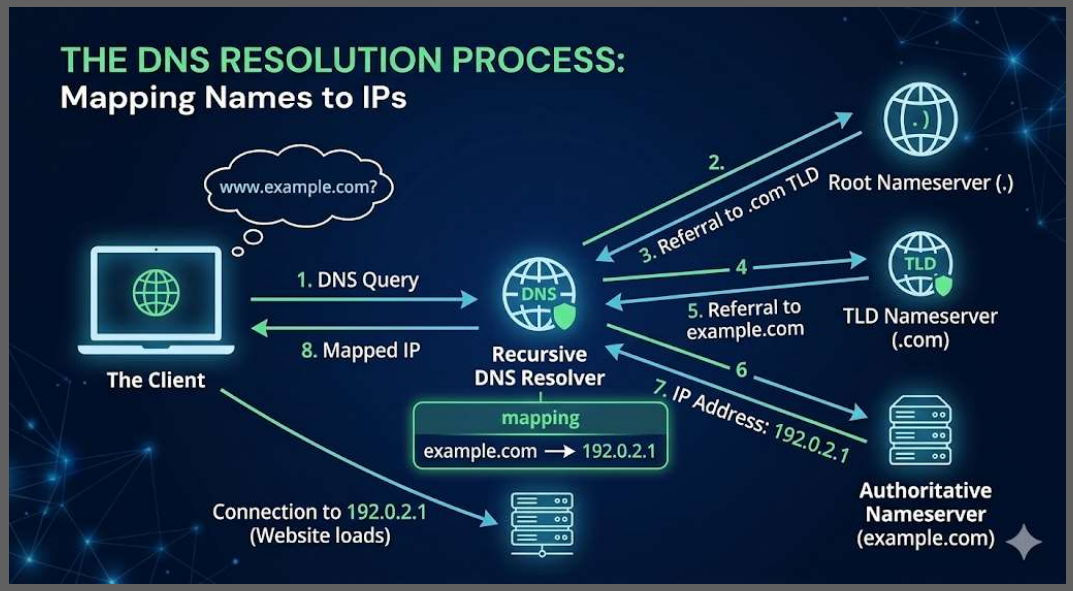
DOMAIN NAME SYSTEM (DNS) ADMINISTRATION

DNS CONCEPTS

- What is DNS?
  - The system that translates human-readable domain names (e.g., dlsu.edu.ph) into machine-readable IP addresses
  - Acts as the "phonebook" of the Internet, critical for locating services on a network
- Key Components:
  - Nameservers: Servers that store DNS records
  - Zones: Distinct portions of the domain name space
  - Records: A Record (IPv4), AAAA Record (IPv6), CNAME (Alias), MX (Mail)

DNS RESOLUTION PROCESS MAPPING A DOMAIN NAME TO AN IP ADDRESS

1. Client Query
2. Recursive Resolver -> Root Nameserver (.)
3. Root referral to .com TLD
4. Resolver -> TLD Nameserver (.com)
5. TLD referral to example.com Authoritative
6. Resolver -> Authoritative Nameserver (example.com)
7. Authoritative server returns IP
8. Resolver maps example.com to 192.0.2.1 and returns it to the client



THE HOSTS FILE (/ETC/HOSTS)

- The "Pre-DNS" System
  - Before DNS servers existed, name resolution was handled by a single text file named `HOSTS.TXT`
  - Today, every operating system still has this file, and it is checked before querying any DNS server
  - Linux Location: `/etc/hosts`
  - Windows Location: `C:\Windows\System32\drivers\etc\hosts`
- How System Admins Use It:
  - Development: Map a domain (e.g., test.example.com) to a dev server IP before the real DNS record exists
  - Local Naming: Give short nicknames to local servers (e.g., `192.168.1.50` backup)
  - Blocking: Map malicious/ad domains to `0.0.0.0 or 127.0.0.1` to block access

DNS IN SYSTEM ADMINISTRATION

- **Role of the SysAdmin**
  - **Centralization:** Managing how computers find each other within the organization's IT infrastructure
  - **Service Provisioning:** Configuring DNS is often a prerequisite for implementing other services like HTTP, SSH, and file sharing
  - **Maintenance:** Ensuring the DNS service is available and records are accurate to prevent service downtime

WHY HOST YOUR OWN DNS? (ENTERPRISE PERSPECTIVE)

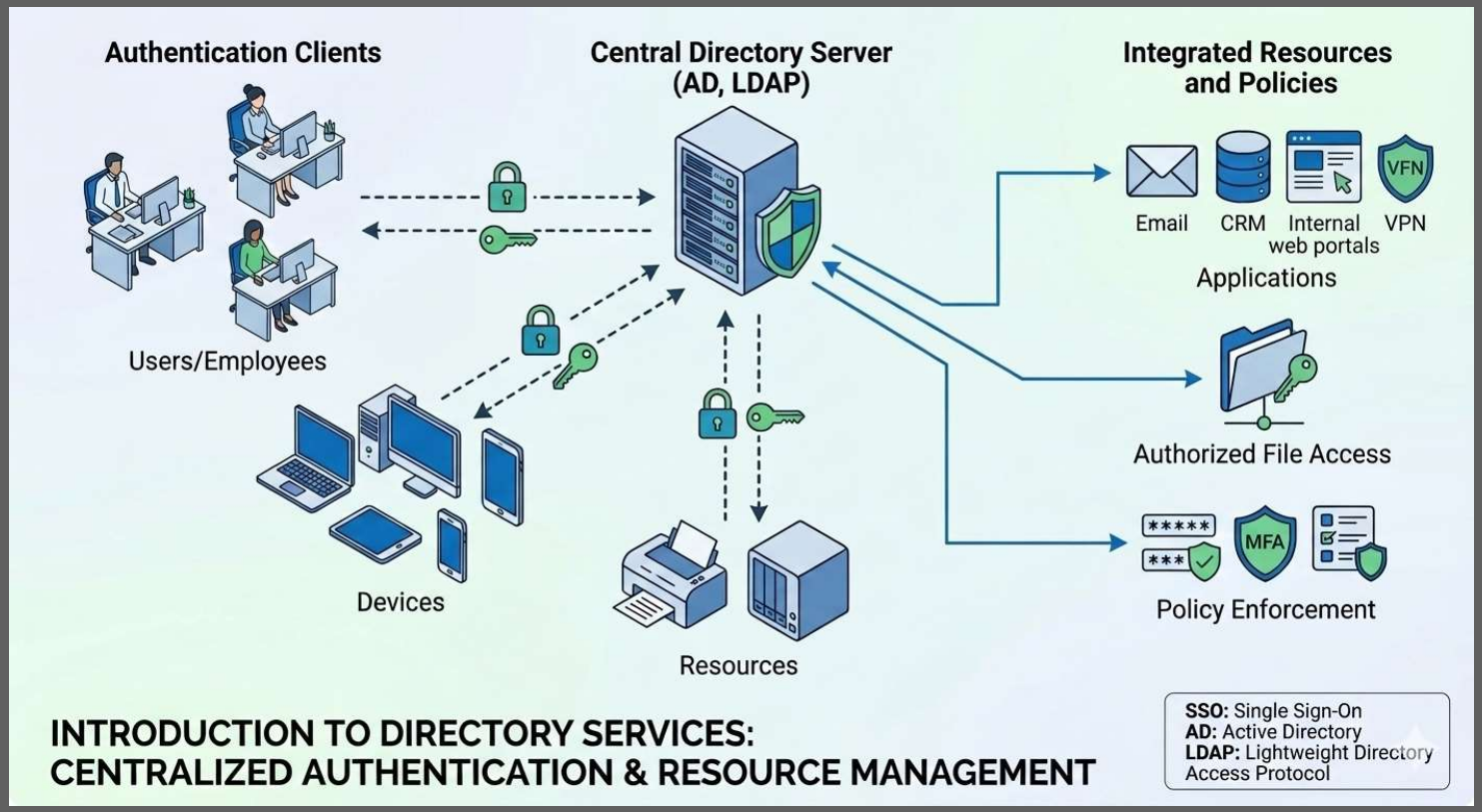
If public DNS (like 8.8.8.8) exists, why do sysadmins build their own?

- **Local Resolution:** Public DNS doesn't know about your internal, private servers (e.g., payroll.dlsu.local or printer-floor3). You need a local DNS server to resolve these internal names
- **Active Directory Integration**
  - In Windows enterprise environments, Active Directory requires DNS to function
  - It uses DNS to locate domain controllers, LDAP services, and Kerberos ticket servers
- **Security & Filtering (DNS Sinkholing):** Sysadmins can configure enterprise DNS to block queries for known malicious domains (malware, phishing), stopping threats at the network level before they reach the user
- **Performance (Caching):** A local DNS server caches the results of external queries
  - If 100 employees visit google.com, the DNS server only asks the internet once, drastically reducing latency and WAN bandwidth

INTRODUCTION TO DIRECTORY SERVICES

- What is a Directory Service?
  - A centralized database system that stores, organizes, and provides access to information about network resources (users, computers, printers, and policies).
  - It acts as the central authority for identity and access management in an organization.
- Key Uses:
  - **Centralized Authentication:** Users log in once (Single Sign-On) and can access authorized resources across the entire network.
  - **Resource Management:** Sysadmins can apply security policies or push software updates to thousands of computers simultaneously.
  - Information Lookup: Acts as a corporate address book for email clients and other applications.
- Common Examples:
  - **Microsoft Active Directory (AD):** The dominant standard for enterprise Windows environments.
  - **OpenLDAP:** A free, open-source directory widely used in Linux/Unix environments.
  - **Cloud Directories:** Microsoft Entra ID (formerly Azure AD), Google Workspace Directory, AWS Directory Service.

DIRECTORY SERVICE ARCHITECTURE



ACTIVE DIRECTORY & DNS (THE WINDOWS ENTERPRISE)

- Why Windows Networks Rely on DNS?
  - Unlike older Windows networks that used WINS/NetBIOS, modern Active Directory (AD) is entirely dependent on DNS to function
  - Without DNS, a Windows domain collapses
- How AD Uses DNS (Service Location):
  - **SRV Records:** AD relies heavily on Service (SRV) records to allow clients to locate critical network services
  - **Finding a Domain Controller:** When a user logs in, their PC queries DNS for the SRV record `_ldap._tcp.dc._msdcs.domain.com` to find the IP of a server that can process the login
  - **Finding the Global Catalog:** DNS helps clients find the Global Catalog server to search for objects (users, printers) across the entire enterprise
  - **Kerberos Authentication:** DNS points clients to the Kerberos Key Distribution Center (KDC) to authenticate securely
- Dynamic Updates: AD DNS allows clients (like a newly booted laptop) to dynamically update their own A records on the DNS server, ensuring the directory is always current

DNS VS. DIRECTORY SERVICES: WHEN TO USE WHICH?

- As a sysadmin, understanding the boundary between Network Infrastructure and Identity Management is important

- Use DNS When:
  - There is a need to translate a name (www.example.com) to an IP address (`192.168.1.10`)
  - There is a to route email traffic (using MX records) to the correct mail server
  - There is a need clients to locate where a specific service (like LDAP or Kerberos) lives on the network
  - Bottom Line: DNS is for finding *where* things are
- Use a Directory Service (AD/LDAP) When:
  - There is a need to verify *who* a user is (Authentication) and check their password
  - There is a need to determine *what* a user is allowed to access (Authorization / Permissions)
  - There is a need to enforce security policies (like password length or blocking USB drives) on employee computers
  - Bottom Line: Directory Services are for managing *who* users are and *what* they can do

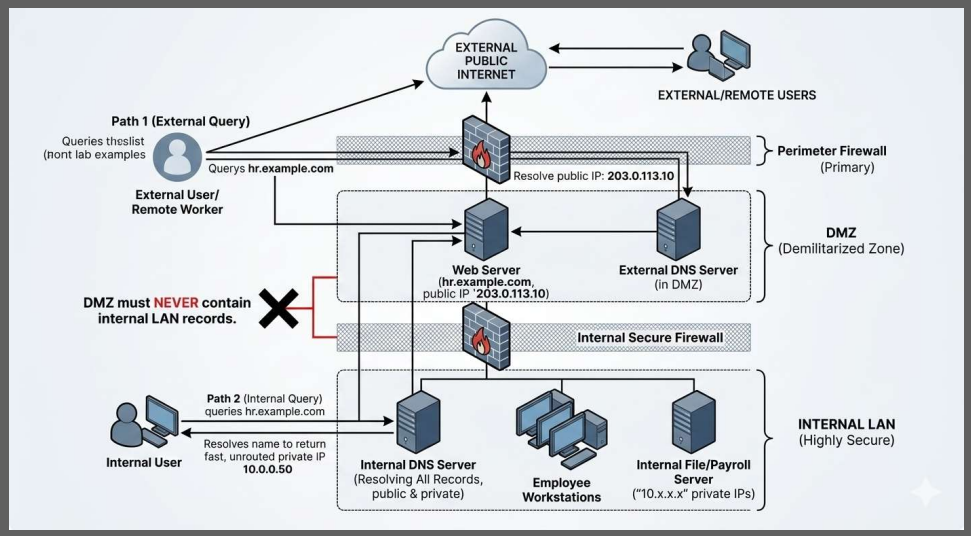
DNS AND DIRECTORY SERVICE COMPARISON

DNS vs. DIRECTORY SERVICES:
Network Infrastructure vs. Identity Management

Feature / Attribute	DNS (Domain Name System)	Directory Services (AD, LDAP, Entra ID)
Main Purpose	Locating Services on a Network (Phonebook for IPs)	Identity & Access Management (Corporate Directory)
Answered Question	"Where is example.com?"	"Who is user.smith and what can they access?"
Bottom Line	For finding WHERE things are.	For managing WHO users are and WHAT they can do.
Core Tasks	• Translate names to IP (www -> 192.168.1.10) • Route mail (MX) • Locate critical network services (LDAP/Kerberos)	• Verify who a user is (Authentication) • Check authorized permissions (Authorization) • Enforce security policies (GPO)
Data Type	IP Addresses, MX, SRV Records	User Profiles, Computer Objects, Security Groups, Policies
Example Task	→ Resolve dlsu.edu.ph to IP	Authenticate Alice.Smith & authorize Payroll access →
Technologies	BIND, Google DNS, Cloudflare DNS	Microsoft Active Directory, OpenLDAP, Microsoft Entra ID

NETWORK ARCHITECTURE: WHERE DOES DNS GO?

- In an enterprise, servers are separated by firewalls into different security zones. DNS must be placed strategically
- **Internal Network (LAN):**
  - Highly secure; where employee workstations and internal servers reside.
  - Hosts the Internal DNS Server
  - Contains records for everything (both public and private IP addresses like 10.x.x.x).
- **DMZ (Demilitarized Zone):**
  - A perimeter network exposed to the public internet but strictly isolated from the internal LAN
  - Hosts the External/Public DNS Server
  - Crucial Rule: It only contains records for public-facing services (like the public web server IP). It must never contain records pointing to the internal network

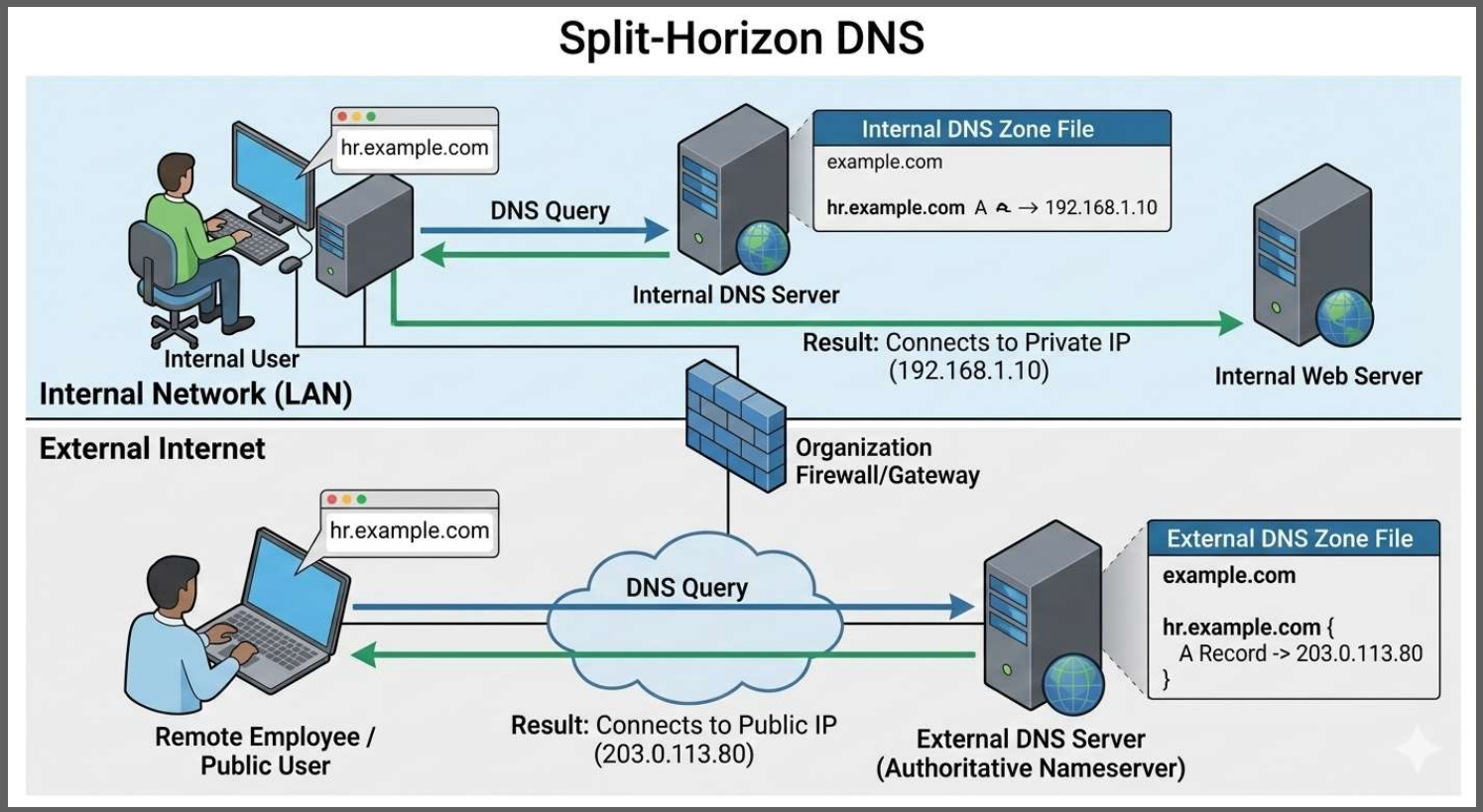


DNS IN THE DATA CENTER (SPLIT-HORIZON DNS)

- Managing Internal vs. External Traffic
- System Administrators often deploy Split-Horizon DNS (or Split-Brain DNS) to provide different answers based on who is asking

- **The Problem:** You have a web server (`hr.example.com`). Internal employees need to access it via its fast, private IP (`10.0.0.5`). Remote workers need to access it via its public-facing IP (`203.0.113.10`).
- **The Solution:**
  - Internal DNS Server: Configured to return the private IP (`10.0.0.5`) to users inside the building.
  - External DNS Server: Configured to return the public IP (`203.0.113.10`) to anyone on the internet.
- **The Result:** Seamless access for users, optimized traffic flow, and hidden internal infrastructure.

SPLIT HORIZON DNS



SPLIT-HORIZON DNS IN ACTION

- Managing Internal vs. External Traffic securely.
- The Scenario: You host a web application (hr.example.com).
- **Internal User (On the LAN):**
  - Asks the Internal DNS Server.
  - Gets the fast, unrouted private IP (10.0.0.50). Traffic stays entirely inside the building.
- **External User (On the Internet):**
  - Asks the External DNS Server located in the DMZ.
  - Gets the public IP (203.0.113.10) which routes through the external firewall.
- **The Security Benefit:** If an attacker compromises the DNS server in the DMZ, they cannot use it to map out your internal network, because those internal records simply don't exist there

PRE-INSTALLATION PREREQUISITES (UBUNTU 22.04 / 24.04)

- 1. Static IP Address Configuration
  - Why? A DNS server must have a fixed address so clients can always find it.
  - Tool: Ubuntu 22.04 and 24.04 use `netplan`
  - File Location: `/etc/netplan/00-installer-config.yaml` (Filename may vary)
  - Key Settings:

```
dhcp4: no
addresses: [192.168.X.X/24]
nameservers: addresses: [127.0.0.1] (The server points to itself).
```

  - Apply Changes: `sudo netplan apply`

INSTALLING BIND9

- 1. Update Repositories
  - Ensure you have the latest package lists to avoid version conflicts.

    ```
    sudo apt update && sudo apt upgrade -y
    ```

- 2. Install the Packages
  - `bind9`: The DNS server software itself.
  - `bind9-utils`: Utilities for testing (like dig).
  - `bind9-doc`: Documentation.

    ```
    sudo apt install bind9 bind9-utils bind9-doc -y
    ```

- 3. Verify Installation
  - Check if the service is active immediately after install:

    ```
    sudo systemctl status named
    ```

FIREWALL CONFIGURATION (UFW)

- Securing the Service
  - Ubuntu comes with UFW (Uncomplicated Firewall). You must allow DNS traffic, or external clients won't be able to query the server.
- Required Ports:
  - Port 53 (UDP): Used for standard DNS queries.
  - Port 53 (TCP): Used for Zone Transfers and large responses.
- Commands:
  - Allow standard DNS traffic: `sudo ufw allow Bind9`
  - OR manually open port 53: `sudo ufw allow 53/tcp and sudo ufw allow 53/udp`
  - Reload to apply changes: `sudo ufw reload`

CONFIGURING BIND9 (GLOBAL & LOCAL)

- Configuration Files:
  - Located in `/etc/bind/`
  - named.conf.options: Global server options (Forwarders, ACLs)
  - named.conf.local: Defines the zones (domains) you are managing
- Defining a Zone:
  - Open `/etc/bind/named.conf.local`:

```
zone "example.com" {
  type master;
  file "/etc/bind/db.example.com";
};
```

  - This tells BIND that you are the Master for example.com and the records are in the file `/etc/bind/db.example.com`.

CONFIGURING THE ZONE: NAMED.CONF.LOCAL

- Declaring the Domain
  - File Location: /etc/bind/named.conf.local
  - This file tells the BIND9 service which domains (zones) it is responsible for and where to find the actual IP mappings.
- Example Configuration:
  - zone "example.com" {
    - type master;
    - file "/etc/bind/db.example.com";
  - };
- Explanation:
  - zone "example.com": Defines the domain name.
  - type master: Declares that this server is the primary authority for the zone.
  - file: Points to the absolute path of the zone file containing the actual DNS records.

ANATOMY OF A ZONE FILE

- The "Database" of DNS
  - Zone files contain the actual mappings (Records)
- Start of Authority (SOA): The mandatory first record. Defines global parameters for the zone (refresh time, expiry, admin email)
- Serial Number: A number inside the SOA record.
  - CRITICAL: You MUST increment this number every time you edit the file.
  - Format: YYYYMMDDXX (e.g., 2026031001)
- NS Record: Specifies the Name Server for this zone
- A Record: Maps a name to an IP

EXAMPLE: CONFIGURING A FORWARD ZONE

- File: `/etc/bind/db.example.com`
- Syntax Rules:
  - Trailing Dot: Fully Qualified Domain Names (FQDN) must end with a dot (e.g., ns1.example.com.).
  - @ Symbol: Represents the origin domain (example.com).

THE ZONE FILE: DB.EXAMPLE.COM

- Populating the Records
  - File Location: /etc/bind/db.example.com (Based on the previous slide's configuration)
  - This file contains the actual mapping of names to IP addresses.
- Example Record Entries:

```
; BIND data file for example.com
$TTL    604800
@       IN      SOA     ns1.example.com. admin.example.com. (
                        2026031001 ; Serial (MUST increment upon edit)
                        604800     ; Refresh
                        86400      ; Retry
                        2419200    ; Expire
                        604800 )   ; Negative Cache TTL
;
@       IN      NS      ns1.example.com.
@       IN      A       192.168.1.10
ns1     IN      A       192.168.1.10
www     IN      A       192.168.1.20
ftp     IN      CNAME   www
```

TROUBLESHOOTING DNS

- Common Tools:
  - `dig` (Domain Information Groper): Query DNS servers for debugging records.
  - `nslookup`: Query Internet name servers interactively.
  - `named-checkconf`: Verifies the syntax of configuration files.
  - `named-checkzone`: Verifies the syntax of zone files. (Usage: `named-checkzone example.com /etc/bind/db.example.com`)
- Diagnosis Steps:
  - 1. Verify the service is running (`systemctl status named`).
  - 2. Check configuration syntax.
  - 3. Test resolution from a client machine.
  - 4. Review system logs (`/var/log/syslog`) for errors.

DOMAIN NAME SYSTEM (DNS) ADMINISTRATION